

Selecting between volatile and non-volatile memories

The most basic division of storage solution for any application is based on its volatility.

On the volatile memory front, where the memory chip is used to store run-time data like stacks and program instructions, SRAM, DRAM and SDRAM are suggested. Access time and cost are the primary factors for choosing RAM, and SRAM is the most widely used for timing-critical, high-speed applications as it is almost four times faster than DRAM. However, DRAM beats SRAM in terms of cost-memory ratio, since these are cheaper for larger memory sizes. It is important to note DRAM does require a controller for interfacing.

Many systems actually use a combination of both SRAM and DRAM to achieve optimum performance levels and minimise cost metrics.

The latest family of high-speed RAMs are SDRAMs, which possess abilities to transfer data at both edges of the clock, that is, DDR and DDR2. These are typically used in media based use-cases involving video processing or in mission-critical systems.

Traditional non-volatile memory options are NAND flash, NOR flash, SD cards and EEPROMs. These flash memories offer block/sector read and write options, whereas EEPROM allows every bit to be modified.

NAND flash, which is commonly used in USB drives and SSD, is slower compared to NOR flash, but has a higher cost-memory ratio and better endurance.

FRAM is the most promising technology for low-power devices. It is a non-volatile memory and yet offers the speed of DRAM. It is 1000 times faster than EEPROM and ultra-low power with 100 trillion write cycles (practically unlimited). Data retention is also guaranteed for at least 100 years at ambient temperature. It is currently offered by companies like TI, Fujitsu, Ramtron and Cypress, and currently supported till few kilobytes, but in the near future, it would be an apt solution for low-power embedded devices meant for IoT solutions.

Most vendors provide capacity information in kilobits and not kilobytes, hence designers have to be careful while choosing parts.

Most embedded applications require both types of memories, as these both serve exclusive purposes.

—Darshan Virupaksha, systems engineer, Altix Innovations and co-organiser of IoTBLR

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field-deployed systems, we need to know what the system needs to do (application) versus what is demanded by the OS. Hence, there is a need to know what is written in storage," explains Sambit Sengupta, demand creation manager, Avnet Electronics Marketing. RTOSes are always very application-optimised and in every embedded design where these are used, these use solid-state drives (SSDs). But in a custom OS, we have to be careful about bad-block management, and hence mostly, these use eMMC based memory.

Compatibility with platforms

Not all storage devices are compatible with all chipsets, field programmable gate arrays (FPGAs) or micro-controller families.

For example, the MT29F family of

NAND flash by Micron is compatible with Freescale's Kinetis K70 microcontroller units (MCUs), TI's Sitara ARM9 processors and Xilinx's Zynq-7000 all-programmable system-on-chips (SoCs), but not suited for Freescale's Kinetis K50 MCUs and Xilinx's Virtex FPGAs. Usually, storage solution vendors provide ready reckoners to check the compatibility of their product with different available platforms.

With higher data width comes faster data-transfer rate, provided there is data-line support available. This means that a memory with 32-bit data width will fetch more data in the same cycle as a memory with 16-bit data width, which will fetch more data than a memory with 8-bit data.

According to Guru Ganesan, managing director, ARM, "There is a lot of developer as well as application migration from 8- and 16-bit MCUs to 32-bit ones today." "The chief reason being that typical modern-day or IoT type of applications require more memory (determined by the address bus size), as well as larger data buses

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